ActInf Livestream #021.1 ~ John Boik

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hello and welcome everyone to actinflab this is active live stream number 21.1 and we're here with john boyk and several other colleagues looking forward to a discussion today on this science-driven societal transformation series and project that we've been exploring and uh had several background sessions dot zero one through dot zero four with john and in the discussion today and next week we're gonna be looking to draw out some threads and include some new perspectives and hear from the questions that people may have and for these discussions we're going to be using the same jamboards that we were using for the discussions with just john and i and we're gonna jump around the jamboards we're gonna hear questions from anyone who wants to ask live or in uh the youtube chat and we'll have fun so we'll just start with introductions and then get a quick little recap and off we go so i'm daniel i'm a researcher in california and i'll pass it first to alex thanks hi everyone i'm alex i'm a researcher in systems management school in moscow russia and i pass it to stephen thank you hi i'm stephen i'm based in toronto i do a lot of work with participatory theatre and community development and i'm researching how social topographies can be used across scales of community psychology and other development practices and i'm going to pass it over to lee thanks stephen uh i'm lee i'm a researcher based at the university of york and i'm studying systems transformation and my supervisor is johan fazee who i think you've drawn a little bit john in a couple of these patients i was really really happy to uh find someone else looking at systems transformation and um active inference at the same time but although i think you're a lot further down the line than i am in thinking about this so really look forward to this session cool well welcome john maybe you could give an introduction and start things off uh sure thanks uh for having me and thanks everybody for joining us uh my name is john boyk i'm a courtesy faculty at oregon state university and the purpose of these talks is to discuss the series of papers that i just recently published in the journal sustainability the title is science driven societal transformation parts one two and three and if you would go we're going to do a quick recap of some of the material that we've discussed before we'll jump around a little bit just to catch some main highlights and uh of course feel free to jump in with any questions as we go so if you could turn to uh jamboard one slide one yep you got it okay good so this just is the pictures of the different papers the links are done on the lower left so in case anyone wants to check out the papers please do they're all open source frequently freely available and we're going to focus today we'll jump around a little bit and capture the main main topics but we're going to focus on part three design today all right slide two please and as as we go through this if there's more questions if someone wants more materials all of the links are available all the links to all the papers and some other materials are available at my website for this project principled societies project and i might mention there's also a interactive model there that you can delete a framework which we'll talk about so you can see how the economic simulation goes uh slide five or slide four excuse me yes so what is the what is the purpose of this what is the what is the problem that this this series is trying to solve um the problem that faces society or really human civilization is a host of very large so far intractable problems including climate change biodiversity laws poverty uh income and wealth inequality you know you know soil soil depletion groundwater depletion pollution i mean there's it's a long list of very serious problems that we face so serious in fact that it's not clear at this point how civilization will fare in the coming decades even in the coming years i mean the world is armed right now the world is armed to its teeth with nuclear weapons so uh you know any moment things could go awry or or things could just deteriorate rapidly as climate change and biodiversity loss gain speed so hopefully we will solve our problems hopefully as societies um hopefully we will improve quality of life and quality of the environment uh but that is the per that is what we're shooting for in the series is pathways to do just that i i would like to point out that this series is focused on transformation in the sense of the novo design of fundamentally new systems societal systems which i talk about six i believe so economic systems legal systems financial systems governance systems things like that so the the the problem is the major uh constellation of problems that we face uh globally and the proposed solution solution is to view society as a cognitive organism and just like any organism it has a intrinsic purpose of maintaining achieving and maintaining vitality so how does it do that how does any organism survive it learns it develops it learns it acts it learns from its actions and and it adapts to new conditions as necessary this series is actually a an r d pro proposal it's it's aimed at the uh the global scientific body and my hope is to interest the scientific world in this in this problem of how do you transform how how could we build a better society what if we were to if we were to create any conceivable way to organize society any conceivable economic systems or governance systems legal systems educational systems what would be the best in terms of quality of life and quality of the environment and long-term sustainability i think that's a question that science is now capable of addressing perhaps this is the first time in history that science had the tools and the skill and the background and the computational power to address these kinds of very very large questions but i think that we can do it now the whole r d program is is is is described as a as a partnership between the global scientific community and local communities so the idea here is to test and to build create and test new systems at the local level in in cooperation with local volunteer communities [Music] um and i and i talk about implementing those systems and implementing field tests via a club model which we'll discuss a little bit more uh okay so that's a very quick overview um maybe on to slide five but first uh stephen with a question and then anyone else who's raising their hands sure just keep checking at the hands yeah go ahead thank you yeah just uh one question i i know she mentioned um architecture so you've got the idea of architecture and you've got these different things that we have got in the constellation and i wonder if you could speak a little bit more to the idea of architecture and how that relates to architectures in the brain or maybe distributed systems of the brain you know yeah you know the brain is what i'm thinking of when i use the work architecture well a little bit i mean people talk about computing frameworks and computing architectures too but i really had in mind the way the brain functions maybe i should make that even more broad because if if we if if if climate change and biodiversity loss and these other major problems that we face are symptoms of a cognitive dysfunction at the societal level i mean that's what i would call it the societal dysfunction cognitive dysfunction because we've known about these problems for decades you know with climate change we've known about it for more than 50 years and yet we have it we've failed to you know adequately address the issues um so i i would say that all of these major problems are symptoms of a cognitive problem or symptoms of a cognitive dysfunction and then the question becomes okay well if we're not cognating correctly what kinds of where do we look for better models where do we look for ideas of how we might cognate better and i would say that in the world of biology including the human brain there's just you know a million a large number of examples that we can learn from like how does nature compute how nature's been at this for a very very long time and how does how does it work in nature and maybe we can uh align ourselves with the same program the same process that's going on all around us and if we do then there's i would say there's the chance of building systems that are much more effective at uh at uh achieving the the uh intrinsic um you know our goal which is to achieve and maintain vitality uh far into the future yes stephen and then and i was just gonna sort of building on that because with active inference one thing that sometimes comes up in uh you know is that rather than the brain being like a computer per se that is a distributed and actually now maybe even not even functionally specialized in certain regions but you know it's more of a distributed system of which you know we're not exactly sure exactly how all old that works but it with the idea of it being a dynamical sort of prediction engine so i'm i'm wondering how that could how much might that inform where you go with this yeah totally because uh i i we're using the word brain but but even when i'm thinking brain i'm thinking whole body and bodies connected to each other other bodies in society that i in in part one of the series the title is called world view and i try to set up a context for all of this and it's part of that context i i i you know i point out that everything is intelligent like every piece of life is connected to every other piece of life you know either strongly or weakly and and the whole whole of life is processing information i mean that's essentially what life is doing is processing information taking reducing the uncertainty about the past or about the future from the past so yes absolutely um you know we have the four e's for example uh and we have uh uh distributed we have like in part three i use the analogy of um between neurons and and individuals in a society and so how do neurons associate with other and disassociate with other there's a kind of a compliment as to how people associate and disassociate with other with others in the in the process of societal cognition and so we have connections in the brain we have connections to our bodies connections to our emotions connections to our feet and fingertips and connections to everything that is happening around us in our in our immediate environments connections to each other and all of that plays a role in cognition okay so i'm not yeah so i'm talking cognition in the large sense not in the narrow sense and i'm certainly not talking about a computer model that tells everybody what to do that's that's not at all what this is about thanks sean lee and then anyone else who raises their hand um it was it's more of an observation really um it's it's just when you were saying that you were drawing on the on the natural biological systems for inspiration kind of occurred to me that then what you're doing is a kind of a a bio mimicry but in but you know building on stephen's point it's not in the older sense of you know the mind is computer but much more in the kind of in your senses you know biology at all levels being able to cognite right right absolutely i think i read by but mr fuller he said something one day that he said i think i'm a process not a thing i'm paraphrasing there but that's right it's like all of life is a process an intelligent process cool and and those and those uh just you know i'll say it again but those ideas really are new to science and and what the i mean you know the mainstream science new to mainstream science in just the last two decades or something like that i mean fairly recent concepts of of uh complex systems that uh you know cognitive systems uh the interconnectedness of all things the intelligence of all things trees communicating with other trees through you know through soil fungus and all of these many of these ideas are actually quite recent in the science world and a funny note on that is that mainstream science is new in the social cognition world the professionalization of science is a new topic and a new genre and i really like some of these points about distributed systems we can think about distributed systems within one level of analysis or one system of interest like within the ant colony it's a distributed system or within the brain it's a distributed system but once you start thinking about brains john you mentioned the 4e the extended inactive embedded and cultured could be more e adjectives but that brings out the richness and says yes it's distributed system amongst the neurons in the brain and then the brain is interfacing with the body and the body with its niche and the tools and the social affordances and it's starting to sound like a lot of topics that we've talked about on active live streams so it's so awesome to to see how we're taking some threads from the past and weaving them together and i hope that we can draw this out how a multi-scale distributed systems perspective on cognition coupled with a social transformation agenda and preference vectors representing policy can get combined through active inference into something that's applied um right right how to act how to make decisions under uncertainty that's what all this all of this is about as a as a counterpoint to some of that or is a different view on some of what we just talked about imagine if we can look at how our society treats individuals i mean we can look at extreme inequalities of wealth and income different levels of access to healthcare you know like there's there's all these issues social issues that we face right now imagine if your brain acted that way somehow you know with with different neurons where some neurons were just simply starved some neurons were excluded from the computational process some neurons were given very few resources just barely enough to survive and others had so many resources that they were they were like wasting their resources because they had so much i mean imagine how well the brain or how poorly the brain might function if that's the way neurons interacted if that was the community if if the community of neurons was anything like the community of individuals in our society we wouldn't be able to make hardly any good decisions we'd be a mess and we are a mess it reminds me of dennett's fame in the brain model and uh the firing rates of neurons are not identical some are firing many times per second others fire not even in a day perhaps so there's there's a lot of distributions in these complex and how do we think about the whole distribution as an outcome of a complex process rather than just trying to identify some sort of an outlier or some sort of single phenomena it's it's uh not the leverage point that system interventions will be effective at so and then any other raised hands this also ties in i think nicely with the idea of post-normal science um the idea that in the post-normal science world where you in high levels of uncertainty you have to make decisions so covid is a classic case right we didn't they had to make choices in january february march april even now and you don't have the data right you don't know and in the same way with active inference the brain and different parts of the body are making action policy selections as is going to inform action and that's not quite the normal realm of where science is you're in this but this is where we're going to be going as we hit climate change and all sorts of other complex questions so that also is quite an interesting tie-in because um kristen i know if you know but he did some modelling on kobit 19 as well yeah he's got into that yeah john maybe you could um unpack that in the context of second order science which you discussed in the papers yeah or yeah or maybe i can just pass it to lee and you can say two words yeah i know i'm happy to um so uh the concept of second order science really is is just that the the idea that the scientists can't be separated from the systems that they are studying so really where that's kind of led to in systems transformation research is is that the only way that we can really understand systems is by engaging with them and and learning how to change them ourselves so we can't kind of stand back and just try and focus on epistemic knowledge alone really and that also implies this kind of second order understanding of ourselves and how our own cognitive systems work as part of the systems that we are engaging with and yeah that was a value-driven value-driven science where the or the scientist is part of the experiment with the stakeholders absolutely while we have you here that paper by faisy was was authored by a very large number of people it seemed and i can you just say if you know the story behind that uh uh well a little bit i think it was authored before i started uh studying with him but um there's a transformation community really i guess of researchers and i think it probably came out of one of the conferences that's been going on so it's a it's sort of the product of the of a discussion between many researchers which is the way that they kind of prefer to write really so again like moving away from this idea of you know the single genius to let's kind of gather and figure this stuff out together right i i reference uh paper uh several times in my part number two so readers can see how that fits into the bigger picture here and it's it's uh funny that that idea that the product of research is collaborative it's a call and a phenotype it's a group phenotype how well this paper is and how many people cite it and how much impact it have it's not going to be reduced back down to a single person and so let's take that and then just like you brought up lee the implications of of learn by doing is that we have to infer by acting basically so then what does it mean for the scientist who's all in on that to be studying climate change on a coast or the conservation of a forest or sociology or anthropology very interesting questions that i hope um a lot of different listeners can see resonating with their own situation so stephen and then any other raised hands i think just also building on the idea of this distributed input is and in many cases it may end up a lot of the may end up that a lot of the work is not in science anymore it's like science informed development or science informed or even science informing a model and the model informs practices which inform the way that communities can get involved and now the real agent is actually the community and we're serving them so this interesting shift between where science fits in and um can change you know i think that's quite interesting as well yeah you know i could have i could have titled the entire series science influenced the transformation because that is the concept i was going for i called it science driven but the the the my my idea there was was similar and also the historical perspective which includes many world knowledge traditions and hundreds and thousands of years and multiple continents not just the story of western academia understanding that that has not been a steady state there's no stationarity with peer review again mainstream science especially the um second half of the 1900s and further that's new so science is in transformation and has always been in dialogue from 10 000 years ago so now given the agency that we have and who we are and our affordances our preferences what's the move so we don't have any sort of a locked in phase to resist against that's not to say there's not inertia but it is to say that there's actually a lot of agility in the system the question is what are the leverage points where we can actually make that impact and how my active inference come into play um you know maybe since we're on the topic maybe we can just switch to us jump ahead to just a little bit if i can find it or you find the slide and then lee go for it go ahead lee just to pick up on your um your point there daniel really i mean um i've kind of been hesitant to say it uh in the actual inference kind of you know discord server in the like but i see such a huge amount of emphasis on active influence as it means to um you know to create ai and to improve ai and i can't help i mean i really spent thinking about how can this knowledge be used to improve you know our own individual cognition and societal cognition human cognition really well there doesn't seem to be um i mean maybe i'm just missing some of the literature on that so if there is any you know point me towards it but then it really seems to be that you know like the whole big assumption is you know we got to put this stuff in ai because that's where all the progress is going to be et cetera et cetera and then i kind of wonder is that really a wise goal i don't know i think active inference and several other paths so not exclusive to active inference it re-thinks artificial part of artificial intelligence because it's part of our extended niche and so yes there's been a lot of focus on applying active inference to machine learning into reinforcement learning those types of areas and then what we're talking towards here is the integration of natural and artificial intelligence under a broader extended cognitive systems scope so humans in the loop or real-time computers just like every stoplight or every bank transaction is going to be every tweet is going to be designing those interfaces so that within each blanket there's functional cognition not viewed only from the outside in but as part of a broader system and maybe we find out that we had good enough artificial intelligence so-called algorithms in and that we just had to hook them up in a different way uh just like connecting different parts of piping could give you an engine that's more or less efficient maybe it wasn't the case that we needed some sort of scaling on a larger data set but we needed a new way to combine sub components which is what biology does as well so just a few more thoughts on that um this the science world and and particularly this this cognitive cognitive framework or concept of what is cognition i think can be useful both at the societal level and at the individual level i mean in the sense that science can help direct the attention of the world to to how how intelligence happens all around us in nature all around us every day how you know how we evolve to to expand on the information process and the information processing the capacity for information processing it helps it help in any you know it just helps an individual to understand a little better or have a different perspective on what am i doing what what is what am i why am i here what am i doing why why am i what am i you know like to to a degree that these this these new developments in cognitive science can help to shed some light on questions like that and at the same time for a society like what are we doing here together what what is our purpose right and if you know your purpose it's a lot easier to know where to go next how to how to frame things in a way that is suitable for your continued well-being and it's a it's a bright line in some senses between first and second order science in the first order science literature there might be a lot of sort of engineering type appeals to well we'll make the society more efficient or science and foreign policy will just simply be better and once we take that second order turn we start to think we're going to have increased clarity about our values it's not going to be an easy conversation we might need to invent new ways of having the conversation or formalizing our values but we need some sort of path that's going to bring us closer to clarity on our values rather than obscuring it with technical advances so stephen john go ahead and then stephen i was just going to say obscuring it with technical uh details but also obscuring it with with incompatibilities between what the science world is saying and doing and what say the governance world is saying and doing or the economic world is saying and doing obviously if all of these systems are part of the cognitive architecture you want them you want to design them so that they are integrated with each other so that the signals you know travel from one to the next just as the signals in your body travel from one you know one system to the next system yep stephen and i think this also ties into partly what this lab has been talking about quite a lot in trying to enable this possibility of making those transitions without losing the rigor of the modeling because the modeling is seen as almost another way to get to empirical data it's not just quite the same as science but the challenge you have as things unlike traditional science going towards engineering you do have this ability to embrace effects and values but the challenge is how to stop things getting prematurely rarefied how it's so easy things get rarified and suddenly effectively you're just using another word for system boundary and you're kind of really doing the same paradigm work with some different window dressing right so it's really that's the challenge i think to to and i don't think anyone's quite managed to do that yet but i think that's definitely we're on the cusp this is this is happening you know right now this this struggle of of what does it mean and how do we do it and the premature ratification as you said stephen it's basically there's a a perfect investing strategy you know if you had the time machine you'd know exactly when to buy the dip sell high we don't have that we're moving forward we don't have access to the future so there's two kinds of confidence errors there would be being overly confident so too early zoning in on a solution that's the premature reification and the other fallacy or the other side of that perfect strategy would be waiting too long and so somewhere between waiting too long and acting too fast is that trade-off space that we have from so many different perspectives with not just finding the value of explore versus exploit that's gonna be best but rethinking the explore and exploit outcomes as part of a process that might be doing something like balancing epistemic and pragmatic value for example in the free energy principle so it's a really awesome um conversation and john continues and altering how you measure fitness relative to those two those two differences well since we've been talking talking about science the role of science in this why don't we jump ahead to slide two uh or uh germ board two slide sixteen if you would yep and while you oh okay that was fast and uh well uh we get that rolling uh just one more thought on second order versus first order science you know fizzy's paper is great really i think an important one but it's nude i mean that's it's new to the science world most of the science world is very focused on what we would call first order science where the scientist is a part is is a is an observer of a system not engaged with transformation of that system or achieving value driven goals in that system and uh first order science is fantastic you know we learn we have learned a tremendous amount of knowledge gained a tremendous amount of knowledge from first order science so it is it is fantastic for what it is and you know has to be continued uh but in addition to that we can also start to be thinking about what is second order science how how can the science world be engaged in a value-driven effort towards some better future and uh i suspect that there are lots of scientists who are frustrated who are frustrated because they don't know how to there's not an avenue for them to participate in in really jumping in to make this world a better place intelligent people look at our world and they see the enormous stress that we're under in this moment and are fearful of the of the even larger stress that's a that is likely about to be flawless so i i think that there might be a tremendous amount of frustration in the science world of how do we how do we use our capacities to actually fundamentally make things better as opposed to just passing off of you know them you know sort of little trivializing it a bit here but just passing off a recommendation to some policy makers who then do whatever they do and you know the scientist is then out of the loop i i i think that this is the time for second order science to flourish this is the this is the opportunity for the science community to really get engaged in these deep and important problems that civilization faces right now and in the in the in the series i'm i'm trying to propose an r d program that is a path for the science world to get involved in these in these deeper questions and have an impact a deeper impact on civilization so this uh this slide is a is kind of a summary of uh from my paper of of why should science you know why should science care about this and why should science get involved and we can just we can talk about any number of these but i'll just sort of jump through a little bit just to highlight a few ideas the very first one is just this idea of societal cognition that's that's you know that's kind of new that's that's uh there's there's great potential for scientific discovery just on that question of how what does a society cognate how does it cognate uh in you know in what ways by what mechanisms by what features and what with what characteristics i mean there's you know just as far as expanding the level of knowledge this question is really interesting um a second one we're talking about designing societal systems and obviously this is a really complex issue these are not this is not a simple how do you build a better can opener right this is a this is difficult and complex issues and the science world has has skills that can be really useful in in addressing these questions and framing the framing the movement towards how do we how do we start how do we test what are the how do we measure fitness how do we what is the goal what is the purpose what are we doing you know like the science world would have a lot to say about uh framing this entire question of what is societal transformation and how would we move how would we move forward with it i might add too to this that as i see it at least one of the benefits of having the science world involved in this question of societal transformation is that the science world can provide kind of object sort of objective and consistent metrics and frameworks to help guide the process turned around uh if if this if this if this process of societal transformation this program of societal transformation was merely based on what a group uh like you know the the this group votes for this let's do this and this other group votes for let's do this and it was just it was just a popularity contest of whatever was most popular in the moment that's where that's where that's where the transformation is going i don't think the science world would have any much interest in that like popularity contests or that kind of uh that kind of effort that where's the sun you know where's the knowledge and that where's the science it would be on the other hand i think it would be very interesting to the science world if transformation was based on some kind of principled approach some kind of some kind like we can agree uh to to an extent at least we can agree what fitness might mean for society and some of that would be based on objective measures some of it might be based on subjective measures but intelligent people can get together and and conceptualize what fitness for a society or for a societal system might look like you know these are of course complicated problems deep problems but if we could do that as a community if we could if we could find some way to assess fitness in a reasonable reasonable way a reasonably coherent way a reasonably consistent way then we have a path to to move forward of you know different different groups could try different ways to reach a high fitness and all of those experiments would be adding to the body of knowledge of how do we do this so that fitness improves right i don't mean to to say that that coming up with fitness scores and fitness evaluations of new systems would be easy but it isn't it is a topic that we can tackle and it's because that because of our capacity to conceptualize fitness and and formalize fitness in the sense of a formal assessment i think that makes the whole proc i'm hoping that makes the whole project interesting to the science world because now we're now this is science now this now this is expanding knowledge this is using uh first principles and you know scientific principles to evaluate systems and and it's not a popularity contest it's not uh you know this is this is adding to our understanding of of of how the world works and how intelligence works and how cognition works you know and expanding it out to the societal level thanks i don't know i mean maybe there's some thoughts on that alone but yep stephen and then anyone else who raises their hands yeah just sort of a couple of questions around that the idea of fitness landscapes and how that might be used is i sense you're you're sort of proposing that you could use these kind of fitness landscape i suppose you could even have descent on free energy landscapes to try and help inform policy selection i.e maybe which research streams you're going to go down um because at the moment you know we're either tied into what the market says or some other methods but this could because this fitness maybe you could separate out because i could see a couple of different ways you could look at that then is it like an extra field of practice to inform choosing the policy of which research field to go down and which avenues or is it also within the field is it with the fields connecting to each other i just wondered how that might be you know it's all of those because because everything that we do ought to be serving our purpose you know i mean i don't obviously there's great room for play and you know and exploration and all that kind of stuff i don't mean to to uh to make humans uh you know two-dimensional or something humans are complex organisms they have complex needs and all of those needs must be addressed right but the fitness i s the concept of fitness i see that i see that as permeating multiple levels right because because yes we would want to choose policies that help us and then and then what does help mean what does it mean how does it look if it helps us what is what is that all about right so in some sense if we just say i want to help we're already thinking i want to improve fitness somehow right but not just policy but even structural design of what how what is a system what what what how do how does society make decisions how do we communicate what is our communication systems how do we get input from everyone into this process of decision making you know like like so across the board both in the structural design of systems and institutions and uh even to some degree um uh uh social norms right through education systems like what kinds of social norms help us um and then obviously too than in policy but then even in our individual lives like it helps i think it might help individuals to understand that what we want is uh improved quality of life lower uncertainty how can i make decisions in my own life that bring me the things that i that i think might actually make me feel better that actually might serve me uh on a on a even a superficial and deep level right thanks john uh lee and then stephen yeah uh just some um thoughts on on the the fitness metrics really so um and this kind of goes to the go to the point about the way that scientists need to engage with local communities because there's quite a lot of work that's already starting in in that area and a lot of it is um it's based around this idea of place scale has already been taken into account and this this whole kind of sense of the emerging bioregional movement which just essentially means you know aligning your political and governance boundaries with the boundaries of ecosystems and that then naturally has the effect of it it partitions things in a way that's meaningful to to local people um and then you can start to ask them you know like what is it that they value within their place because they're already kind of connected to it in a way um uh and i mean i suppose one thing for me is this idea of metrics really versus counter factual simulation because once you start putting a metric on something you know like you have a number and then you know that's something that people can't really care about whereas you know like if they can imagine how their you know their local river system for example uh wouldn't be polluted by you know sewage being released from the sewage works or you know run off from them you know and they could see you know that then has this sort of the effect of um you know of engaging people i think more in in the uh in the actions that we would need to collaboratively take to make these kind of things happen sure sure other different people might react in different ways to some people might be more you know really appreciate the modeling aspect other people might just appreciate the boots on the ground local change aspect but i would say there's room for all of that and uh the bio regionalism concepts are great i've been a fan for a long time and who knows maybe in the future we'll move more and more towards that so that communities are really localized in their environment and and protecting their environment and all that kind of stuff that's great in in some ways in some ways science has to catch up with communities of change who have been working on this for a long time these these concepts and in other ways science is already you know pulling pulling communities you know helping to save more there's more there's more there's more we can do there's a bigger picture we can serve here so it's a it'll be a give and take i think between communities and there's one more thing is the bioregionalism idea is great and a lot of work also has been done uh in alternatives to gdp so you know the metrics that are that people are talking about about you know considering education and longevity and disease and all these other things as as opposed to just a single gdp measure of societal quality so there's lots of work being done on multiple fields and to speak to the metrics from an active inference perspective let's just say that we wanted a river that was uh unpolluted we wanted no toxins in the river so our preference in the active inference model would be that the river is clean that'd be in our preference vector now traditional reinforcement learning would look at the level and it would reinforce policies that reduce pollution locally and then it would forget policies that didn't locally now the question is let's just say you start decreasing it and all of a sudden it starts creeping back up your reinforcement learner is going to be quite confused so an active inference we can say that our metric is actually our uncertainty around our generative model of the river's pollution so then what actions can we take if our uncertainty starts to get wider how can we increase our precision on our generative model of pollution and then of course concordantly with our values that it be unpolluted how can we act to reduce our uncertainty and so it's the reinforcement economic maximization mindset pick the metric i want everyone to read that's the metric reading well no the preference is that everyone can read but the metric in active inference is our uncertainty around our observations and it's an extremely subtle at first but eventually very far-reaching way to think about these systems and what fitness is because a lot of times fitness metrics people are going to be thinking right like gdp or write like something that we want to see but we can actually now tease apart our preferences from our affordances from our uncertainty in our generative model and several other features of active inference scale-free models and that is going to help us have a multi-stakeholder conversation in a new way that's great great great great great great great thank you um yeah i just add to that i totally i totally agree with what daniel just mentioned there because it's not almost it's not an active inference per se on its own but it's the ability of active inference to be used in this epistemic foraging way and i think the point that was made there about reinforcement learning which we kind of think oh that's some machine learning method but actually reinforcement learning is the dominant approach to development it's like it's the way humans have structured in a way their own thinking it's not exactly reinforcement learning such a bit but it's it's so constrained and it's so goal driven um and the only integrating way to overcome it and why thatcher then started saying there is no alternative is the market saying it can't deal with the complexity beyond a certain point so you have to offload it to this market which again you get into that kind of just becomes whatever's the favorite of the day this gives us a way to get into this nuanced complexity it gives us at least the chance to go somewhere other routes and plausibly deliver on that promise you know i think that's really cool it's to give like a spatial metaphor on reinforcement learning versus active inference it's like if there's a mountain peak that we know that we want to get to the reinforcement learner would say okay the goal and the reward is the elevation and so it can get trapped on a local elevation peak but then it might not take a step backwards or down a few to get to the highest peak but again if we do a trajectory of policy inference on getting to the top of a mountain again of course as a preference that we have for ourselves in the future we can ask am i reducing my uncertainty about how i'm going to get there when i'm going to get there and the way that we're going to do it together with potentially a group of hikers who have different capacities instead of how are we going to get to the top of a little null and then just stake out so it's a different conversation style steven yeah than anyone else yeah and so building on that is when you have that come into play is the ability for the active inference approaches like you're talking about is magnified greatly when you don't just focus on the one person or the two people who are like the agents of change but you say well what about the local people who have no idea about the goal they have no ability to elevate they have no ability to contribute to the reinforcement learning but they know the slopes they can tell me the history they can tell me where people fell to their death in the past or where there's you know all these and that then makes the place made place-based community-based contributions much more plausible um as why they they enable the scientists and other people to do exactly what you're talking about excellent uh just uh just to point out so we're talking about reducing expected uncertainty about the future right think of all the ways that that think of all the implications of that in say design of systems or design of institutions or actions of humans you know facing some problem right so we so one question that immediately arises is do we have enough information are we collecting the right information at the right frequency to even understand what's going on or to reduce our uncertainty second is how good are we at anticipating the future are we do can can we do that with some accuracy do we know what's going to happen next we have any idea what's going to happen next how good or how good are our are how good are our worldview concepts that lead to anticipation of the future and how good are our you know computational models about what's going to happen next how good is our information information sharing capacities how are we communicating with each other are we communicating the information between groups and between individuals such that we can understand what's going to happen next and understand what our uncertainty levels are you know like i can go on but there's like this it seems like a simple proposition at first and then you realize that proposition has enormous depth it implies all kinds of things about how a society might function and organize itself one one thought on the wisdom of crowds and distributed computation there's a famous example from early in quantitative genetics where it was at a world's fair and a crowd was asked you know you put in a raffle ticket you can predict how much some cow uh weighs and then the average of the estimates was extremely accurate although many individuals were of course distantly off the mark and a lot of people take that just um straightforwardly as existence of the wisdom of crowds but there's a few pieces of that scenario that really matter the first is that the people attending may have had experience they may have been working in agricultural settings or have experience with getting feedback on their predictions and secondly the estimate was a bit bounded it wasn't that the cow was one pound or ten thousand pounds there was a an estimator and then it was still a cognitive diversity that contributed to that estimate so you couldn't just look at somebody who overestimates that well get rid of that guy he's wrong he's like that's counterbalanced by someone on the other side of that curve that's how we're getting an estimate so having a distributional and a community perspective on decision making helps us recognize and encourage and foster diversity of opinion and also structure a resilient community so it's something that will be really interesting to see how and where these kinds of memes and ideas pop up yeah absolutely um valuing differences of opinion valuing differences of perspective is one one thing that you had said there i just added just add something to that is uh in our polarized society today we we you know there's like liberals and there's conservatives at least in the u.s they seem to be in two different universes almost right but liberal and conservative at their best is a little bit like exploration versus exploitation you know the the necessary balance for any organism or for any decision-making process to balance exploration of the unknown and exploitation of what is already known what is traditional what what is our history what what do we know works uh you know like at their best liberal and conservatives are both important just as exploration and exploitation are both important and problem solving and they're relative just like left and when you're spinning around and things are changing it's hard to say and that's why we can pull back to some real pillars of biological systems instead of trying to scaffold further and further into the political or logistical ether we can actually return to our grounding as a biological society so lee i saw your hand raised or maybe i didn't or stephen oh yeah stephen and then anyone else who raised their hands that was quite interesting we made that point though about you know we'd look to explore the unknown and i think that in addition to that um because that's from the perspective of say the scientist or the person in the kind of the knowledge um generating game so to speak that's their profession or whatever but there's also that exploring the known but not knowing that it's known the tacit the so this is where communities can come in is like there's there's like they i realizing that some of that unknown is participants in those environments no it just needs to be surfaced this is where some of the work with the environment side of the active inference the niche and these topographies that i'm quite interested in my side can come in because now they're exploring and revealing stuff in a way which is networked and sort of not just isolated as single stories or experiences and fragmented so i i think that connects with what you're saying can you give one example of what you're thinking so one example we did a project around water issues in rural areas of northern maputo land in in south africa and so they had different areas of water catchment so some people were getting water from a lake but the lake had hippos and you know crocodiles so it was a bit challenging for them in different ways and um so you've got these different water catchment areas and it's they had the lived experience of having to go there having to get the water having to boil it because if the water's contaminated you've got to boil it and there's other um what it means then if you're then doing that if you've got someone at home who's got hiv on top of that so then you know so they can't walk or they've got to have so all these questions so what they would do is they would create a fabric map the size of the room about all their different routes to water and then share the stories from eight different or whatever numbers of hot spots and then that became the sort of landscape so in doing so that that wasn't given the answers but it was informing another process maybe a week later which was looking for innovations looking for options for new approaches so that's one type of way that you could like reveal something from the local landscape which would never have been available to an engineer from their own knowledge yeah great stuff great stuff um so some of what you're talking about is what is our narrative what is our individual narrative and what is our shared narrative right like what what is our life what is our experience of life and what does what do i as an individual have what what knowledge and lived experience can i offer to this group that would help the group to eventually function better or make better decisions right that's a really interesting question to me for many reasons but one reason is how suppose we were to move forward with this this whole project was moving forward how do we communicate with each other richly how what the i would i would argue that there is no system no functional system right now that a million people or 10 000 people can richly communicate their lived experience their their understanding of life their understanding of how what might happen if a b and c happen i don't think we have we're in need of tools to richly communicate uh uh our you know i mean we're needed tools yeah so that's a immediate opportunity for groups to get involved with is just at just that little question of how would we communicate and how would we how would we how would we make decisions together in some kind of collaborative fashion probably how do 10 000 people do that once or a million people when you can't all fit in a room you know that's the question when you can't all fit in a room to really thoroughly discuss these issues and listen to each other how do we do that at scale right so that is i think there could be answers to that that's not an impossible task that's a really uh interesting and doable task if we just put our minds to it it's but okay let's relate to regimes of attention and active inference perspectives on narratives which were even our first active livestream number one was on narrative active inference because we think it's crucial and if there's going to be a social narrative like you had brought up well that's something that is not just sharded into a person's brain it's not a holograph where it's the same in the person's brain as it is at the social level things that brains know that neurons don't it's not like each neuron is holding a little mini brain it's actually that there is differences amongst neurons or if you think about like star wars or movie it's like each character doesn't know the whole omniscient scope and so we're part of a bigger story and so it's going to be interesting to explore what will it feel like as a person and what will it be like experientially with a personal narrative to be part of a broader societal narrative but those will not be the same thing and the desire for those to be the same thing will lead to system fragility in many cases so stephen and then any other comments yeah i think that that that ability to know certain things is is a very important point and i think in active inference something that i've increasingly been looking at we've got external states and we've got hidden states okay in different ways so there's from the perspective of the generative model of a say a water engineer there will be more like there's hidden states from everything right from everyone to some extent okay as well as states that we can engage in um and so there will be this question about how do you engage the hidden states in a landscape as well as the external states of the regime of attention and where can someone else who's got a different lived experience contribute to opening up and maybe telling us where they think the meaningful aspects of their landscape are so that i think that as well as modeling happening from our own computational models and being able to maybe use some tools that we can model through participation in placements place basically absolutely absolutely so so this tool i have in mind it's uh it's i want to have my voice heard i i have a story that is that could be important to others that could be important to any particular situation that or topic that society is trying to decide upon and i would very much like my voice to be heard and not just a yes or no or not just to choose a b or c way i would i have a rich experience that i want to communicate you know i another person sitting next to me might have a model of water quality or whatever the issue is and they might have something to say about what their model implies but all of those voices right should be heard all of those voices should be part of the cognitive process of a society so so certainly data driven in you know model formal models of this and that sure that's great that's fantastic it's an extension of our cognitive capacity and at the same time i'm intelligent and i'm you know i can do things a computer can't at least today you know i can do many things a computer can't so whatever process whatever communication system and process that develops out of this ought to be one where every voice can be heard can be registered can be incorporated into this larger really cognitive process you're talking about you know it's like sharing and we're discussing but that's a cognitive process right for a group because it's leading towards some better understanding of the world and you know improved decision making about the world john if i could uh you mentioned data driven so to again sort of distinguish and contrast with how you're potentially using data driven versus how it might be uh implicitly used in other settings data driven who's being driven us the data are driving us and our policy oh the policy is going to be data driven okay well how about it's human driven how about the policy and our affordances and the data are like a car and we're driving the car do we need a better car does it have too many wheels where are we driving who whose turn is it to drive how are we choosing which music is being played in our human driven car not how are we going to fit into a data-driven human mode and it's just interesting that sometimes what is left unsaid or unstated is where a lot of those values get swept under the rug it's data driven what else do you want to talk about oh values okay well there's time for values it's like there has to be time for values but the data-driven conversation makes it sound like it's the beginning and the end right right today yes yes and i think that can be expanded greatly right so that's that's when i use the word richness that's a little bit of what i'm trying to imply is the richness of experience the richness of perspectives the richness of of past information that cannot that i as an individual can bring to bear on this issue whatever the issue is and then on top of that what what does the data what data do we have how frequent how frequently do we collect it how what is its quality how can it contribute to this discussion right so very much it's not that the computer has to decide what to do it's that human beings have to decide you know what to do how to given all this information what do we do how do we frame this how do we frame our problem how do we are we even looking at the right problem are we look are we is our focus too narrow are we you know or do we need to broaden our focus to understand better how all of these different problems so very much human and the human in the center computation and to connect that to first and higher order science and i'd be curious what you think the first order project is like build a bridge the second order project is like given what needs to happen is a bridge the best solution or maybe we can go about it a different way and the third order is well what really needs to be done here what do we value who are we those types of third-order questions which potentially even um are on an escape trajectory from science because we're not going to be looking to science in a narrow sense for the answer to that third order truly human question but we do need to expand this discussion from what is the best way to build the bridge how are we going to go about fulfilling transporting people from side to side or making a landmark for our city there's different reasons why we might want to build a bridge and people feel like a cog in the machine if they're just being instructed at the first order like do this right but when we're including people in higher orders then we're part of a multi-scale cognitive system we're not just a quote cog in a machine so then anyone else yeah i think this um i think it's a really important point and for me it touches on um you know metzinger's uh transparency opacity kind of uh you know are we do we understand the the beliefs and the frames consciously that we're operating out of so in the in the bridge metaphor you know you might say um you know at the third level the third order oh it's it's really about how are we looking how you know what's the ontology or the set of metaphors that we've even coming at this system through and what effect are they having in shaping uh you know our well a how we perceive it and be up then our actions towards it so you know going right back to the beginning of this discussion it was quite interesting when stephen asked you john about architecture you know and i'm kind of thinking is you know is that um you know is that really is that a metaphor you know the architecture of the brain is that is that a way of looking at a brain or is it something you know this parallels the discussions about markov blankets etc etc you know like are these things that we're imposing on systems or are they things that are really there um i think the architecture one is really interesting because it kind of comes from this you know whole constructivist tradition and i mean i wonder more whether it's not so much whether it's about the right you know the most accurate way of seeing but what particular sets of affordances does a model or a way of seeing offers you know in the face of this particular problem in this particular niche perhaps so yeah interesting stuff thank you lee um john continue or stephen and then john feel free to continue yeah just one last piece i think that this there's some areas that i'm trying to apply these isms now and to sort of put some buckets i think the social constructivism is is an important one to think about maybe science in a way kind of fits in there in a way as a human pursuit pragmatism i think is very well aligned um with approach of active inference and this approach here if you take pragmatism from a you know a human not a humanistic perspective well from the third one which is an ecological perspective so or any ways of knowing as well not even perspective so you can drop down into ways of knowing perspectives you know pragmatism social constructivism and ecological approaches and i think nexus is is quite important as as a sort of way to hold some of these ideas this is you know this is a fantastic conversation and uh and i would say that um exactly the kind of conversation that would be needed to to move such a project forward it would be conversations like this you know between scientists within the communities between communities and touching it on the all these uh you know some some superficial and some very deep and subtle issues and questions how do we what are we thinking how do we do this how do we how do we measure success what do we want what is our purpose i mean you know we're just maybe starting that conversation here amongst us but i would like to offer to the science world that this could be we know this could be a function in the science world of uh you know a pursuit of study a pursuit of of uh attention and uh if it were i think that the you know the i think that's all that's really necessary actually is to put our minds to this and intelligent people uh communicating amongst themselves and engaging the all interested stakeholders could definitely make progress in this in this endeavor so great great to watch great to be part of yeah john maybe if you want to go to another slide but that's leaves the question how do we in our dot one video it's kind of an opening the dot one is where we've had our background with the dot zeros we've hopefully read the papers and digested them written comments written thoughts to ourselves allowed that to reverberate and then we open the doors and we come together in a group discussion for point one and we also say hey the doors aren't closed more people can join this conversation whether it's next week for point two or on a broader scale so i'm just wondering whether you answer now or later who can be included and how can they be included in participating a few words um everyone is everyone is welcome in this conversation uh in in the in the world and communities in different uh in different different types of communities the science community other communities i mean this is this is a conversation for all of us to be had what do we want what is our purpose what are we doing how can we how can we improve uh how can we reduce uncertainty uh you know about essential variables you know to put a check more technically how to do that i don't know i i'm open to suggestions of how we make the next step but you know i sort of put out put out what i could put out in this series and i'm hoping that these conversations might uh lead to some actions that uh engage a larger and larger sphere of of individuals and organizations so i'm open if if there's suggestions i'm open um could you flip to uh let's see that would be slide two uh talk to slide eight yep go for it we've touched on all these ideas and i thought maybe this might be a a good moment to to talk about some of the practical implications when in our original uh discussions this slide was coming after the after paper one which was all about world view and and seeing life as a cognitive process seeing individuals as as overlapping uh processes of information flow and understanding that that when we talk a a community improving its its quality of life and quality and environment we're talking about an extended individual an extended community that really reaches all the way to the biosphere ultimately right we're all connected with other individuals and all communities are connected with other communities and all of life is connected we're connected with our local ecosystem and with this new ecosystem so when i when i talk about uh vitality i'm talking about the vitality of the whole really that's our purpose is of human we're intelligent we're humans are very intelligent we can think a thousand years into the future of what might happen if we do something today and and humans can use that capacity to improve and uh reduce the uncertainty you know of the whole quality of life can prove the quality of the environment and biodiversity and all that kind of stuff so uh with that kind of theory uh theoretical aspects in the past what are some of the practical implications like how how would we use any of that information to actually think about design for societal systems right so um so this is a list of something some items that we have talked about in previous talks and even touched on today and when we think about what does this mean practically i would like to focus that to a different question really and that is how do we how does any individual any organism solve a problem like what is necessary to solve a problem whether you're a bacteria or you're a human's community right what do you do and and on the left are i would say some of the universal aspects of of cognition uh for example you have to gather data i mean you're the intelligent being is processing information and information means that some kind of data information has to come from the world and then it can be processed it has to remember for example it has to anticipate the future forecast the future with some accuracy in order to survive if if i see a you know if a lie if i see a lion in my path uh i might want to accurately forecast that if i don't get out of the way i might be the lion's dinner right i mean anticipation is central to cognition so so we i've only talked about the first three in the list there but already let's we could talk about what does that mean for a community then so how does a community gather data for example and that might be you know through sensors through through getting the input of others in the community about some particular issue requires remember like remembering requires that that information be kept in some kind of accessible form some kind of repository of information we can think already of ways that society does the does these now we we for example the census bureau collects information uh you know annually and every 10 years collects various kinds of there's there's financial surveys done every few years uh some of that information is put in repositories that that at least researchers can access for different reasons but how you know how does that how would that differ how do how does what we do today how might that differ from if we were really thinking of communities as cognitive organisms how might we improve on any of these processes right and i don't know if there's thoughts on that but i can yeah yeah stephen stephen go ahead yeah i mean one thing i think that's interesting is this this actually points as well though to an interesting dilemma i don't know if it's a dilemma but certainly a challenge in the sense that um these artifacts and these artifacts of knowledge that we have they're kind of we have the option in a community setting to place them in our environment because they aren't they can be equilibrium based artifacts so for instance if i create a book and i put it on a shelf it could be their hundred years right and they'll come back and it's still a book right so you've got this ability for that kind of storage now when i remember something i i'm with active influence i'm establishing this kind of priors an update of priors um this is generated on these lower level sensory flux pieces of um sensorum which i have at higher levels potentially started to interpret as what you might call a signal but it you get into this interesting question because it's it's like the error is kind of the signal in active inference although it's possibly at higher levels of thinking it gets a little bit different because we've been talking about that with some of the modelling but there's a there's an interesting question around as you scale out where some of this sense data in active inference is kind of held in the kind of collective knowing of a which knows how to interact for instance with the artifacts yet you've also got aspects a bit like i've got my clothes and i've got other things that kind of they part of me are they not part of me they're part of the environment you you there's i think there's something quite interesting and i i think that's an area i'm really interested in exploring is is teasing that out so i wonder what your thoughts are on that well part of part of i think part of what you're speaking about might be what i've been calling richness of information right so so obviously a a census can collect information once a year about you know numbers and education levels and things like that that's one that's one stream of information a valid stream of information and yet another is very personal it's what it what is your gut you know what is your gut telling you about this situation or this you know this this constellation of what's happening today what is your gut feeling of what's happening right so how does the how does a community get you know process uh sense and process all of that information together if you have a you know if we're if we're part of the community and we're trying to decide on some issue it would be good if we heard your gut you what is your gut telling you about this is this good or bad or should is this direction good or bad or you know even if even if there's no like data data to back it up what is your sense based on you know the the the your your complexity your wholeness your like we can't even put a word on it what you what you are or where that information comes from maybe but what is this what is the sense that you're having and i would like to share my sentence too and everyone maybe could share their sense so how do we how do we do that in how do we do that when there's a hundred thousand of us how do we do that uh to give just a few thoughts there's there's two sides to the conversation they're sharing communicating and then there's really engaging and listening which is part of the idea of attention topic so 100 000 people speaking over each other is zero conversation that's not yeah yeah that's not conversation right exactly so how do we how do we do and and i don't think that that's not you know like it's almost surprising in a way that that we we have such enormous technical capacity in today's society to do unbelievable things and yet this very basic question of how do we as a community of say 10 000 or 100 000 how do we richly share information in a way that all of us feels heard that that are that we can express our our our kind of like hard knowledge and gut knowledge how do we participate together in describing our situation describing our fears describing our uncertainty anticipating what is going to happen next if we take choice a b or c you know you it's kind of interesting that that that that that tool doesn't exist in spite of our technological process and there might be reasons why that i mean there might be political reasons and economic reasons why that tool doesn't exist because if that tool did exist people would want to participate you know like naturally i want my story heard i want i want my community to understand what i know because maybe it's going to help somebody and and i want to hear what what you know in a in a healthy conversation in a healthy exchange i really want to know what you're thinking and i want to i want to contextualize all this information in light of what our purpose is what our goal is what are we after what do we want well can i comment on that because i think this is a really important point and and also though as well as see this is what i find i'm i'm recasting speech as an action policy as an extended cognition into a spatial landscape of semantics right so effectively it's actually words could be seen as in a virtual environment that we're projecting out and what that can kind of open up is when i'm talking to someone and being heard right which basically means being words are being heard right so but there's there's certain things i can't communicate through words because they're not in that semantic landscape which is the compression of my a useful tool but the thing with tools generally speaking it's not very healthy to relate to tools right it's like tools of things so one non-verbal work can be really powerful we actually had that with our work within the community who use alternative communication you know we've really pushed that so they it can take them three minutes to do three words because if they're using the head switch and they have to scan through each letter and then hit the head switch and so we'll really about how about it not being about the words and about being the look of the eye and that building up of non-verbal knowing between people which at some point will probably be well we actually then articulate that through fabric landscapes so then they start painting the room and do world building and then you start making the words so there's there's something about um like you said there may be something that taught well we need we need the tools but also sometimes tools just can't go there so what do we do where we don't use tools um and then when do we bring it into a tool-based environment i think this is a good point yeah yeah yeah and i think much of human communication actually happens non-verbally too if we can be in the same room and i'm watching your facial expressions i'm getting a lot more information about you and your intent than if we if i was only hearing your voice if i was only reading your words so sure there's there's nonverbal communication there's artistic expression there's there's poetic expression you know using words in a different way obviously we again we as an individual i want to be heard and if i'm maybe i'm expressing myself in words maybe i'm expressing myself in my how i'm raising my eyebrows or maybe i'm express maybe i don't even have words to express myself but i have an image in my head that i want to communicate somehow somehow we we have to be able to communicate richly in in mass and how do we do that and part of the answer to that might be we communicate in smaller groups and that information then is is sort of propagated up the hierarchy or something maybe i mean you know the the the the slate is wide open here so this is the question to the world how can we communicate richly in mass such that our purpose might be fulfilled you know our our purpose of of maintaining achieving and maintaining uh vitality into the future right so that's the big picture and then how what are the what are the what are the many tools or social norms or educational processes or any number of things that lead us to fulfilling our purpose you know our intrinsic purpose i should say you know our biological purpose in a sense nice john i hope that people clip that out or hear your question and really consider it and what it makes me think about on this slide and to return it to active inference again is really like the rhetoric of participation because in a reward maximization culture things like communicate it's like but not now we have decisions to make whereas when we reframe about reducing uncertainty on a multi-scale so individual and shared generative models it's like have we communicated enough to even know each other's generative model because once i know that your generative model includes this dietary restriction i don't need 100 text messages a day i actually can generate recipes that are going to be compatible with you so it's not like we just need to have more communication or a different kind or these two people need to talk or some sort of secret trick it's actually that we can balance a lot of these sometimes locally contradictory needs choose two and you can find a situation where they align on this list and you could find a situation or imagine a situation where they diverge and the power comes from being able to have that goal directness and the epistemic gain and then potentially through peer mentorship or through extended cognitive architecture we could be able to mark you know where is there a yellow flag where is there a red flag where do we need to pull back and share more um about one person's generative model how are we working as part of an integrated unit that's reducing uncertainty given the preference vector rather than well if we each just make as much money as possible then the group will have as much money as possible right so this is a different way that we can go about having that conversation and as hopefully we're drawing out it's something where everyone's perspective uh isn't just a uh group to be formed their perspective is their contribution and their value yep yeah good good we can also look to biology too how does how does uh the cells of a body make decisions you know like how does a liver know what to do and when when to act how does the heartbeat know when to speed up or slow down there's rich communication that happens multi-scale some local some distant through hormones through messengers there's enormously rich communication that happens between tissues and cells in order for all the pieces to understand what the you know what what what is necessary now right so i mean there's we can we can this is an open question it's a clean slate there's a there's a great potential for the science world and other communities to move forward of this question of how do we richly communicate and make decisions together how do we do that and uh and there's lots of biological uh examples for us to look at to gain inspiration from so a question a question from the chat just before stephen um from yvonne uh just about the tools how many people have used the kinds of tools that you're suggesting or it sounded like you suggested that it didn't exist but what is the preferred field for a startup from yvonne where does this tools where is this coming out of this sort of tool for communication collaboration sure i understand the question exactly where is it coming from he asked what is the preferred field to make a startup in for this kind of a tool like would it be in the field of industrial engineering or oh so i have ideas in my head of how we might you know start to build such tools the you know the reason i'm thinking about it is for this very this very problem of how to communicate how do communities communicate how if this club model were to if we were to move forward with uh with studying this club model and eventually developing a field trial with a club model how does communication happen within that club model so that's that's my interest you know is the application to this problem of societal transformation and and demonstrating new ways of of cognition at the local level via clubs but obviously if you had a tool that could do this there would be tremendous opportunities for others to to use the tool too this is just one example of you know thousands you might think of um i used i work in the utility industry and i've seen how messages get transferred from you know a caller calls in and says you know there was a there's a there's an accident there's a wire down there's sparks flying there's children around you know there's some kind of rich situation happening that others need to know about and by the time that message gets to the person in the field or in the truck uh you know it might be quite garbled it might be it might be not unclear at all what the problem is or what the dangers are or what the situation is right so that would just be one example of how industry might use a tool for better communication and and you know there would be many many more examples you could think of so i would my guess would be that if the science community moved forward with this problem of how do you richly communicate with a large group those tools and the concepts that would come out of that could be applicable to a to a wide variety of industries and to this particular question of how does the club model work and how do how do communities cognite thank you john stephen and then anyone else with raised hands yeah thanks i suppose this sort of reminds me of some of the principles actually systemic coaching can be an interesting area like looking at coaching groups that and the system it seems to be that um and i was just wondering if you've encountered um the viable systems model stafford beers viable systems model because i think that some of the active inference work can recast that you know because you've got not just levels of regulation on the on the body because it's based on the i think it's the body on the nervous system of a frog um or um or different nervous systems but that doesn't necessarily have that bottom-up teleology that you get from active inference doesn't it so you have the kind of the idea that the lower levels also know something it tends to be the higher levels of regulating the lower level so but uh hey that might be something to look at as one area that'd be interesting and um yeah i just thought i'd mention that in case you know no no great uh and uh you know those are the kinds of ideas that would you would want to explore as you build the kind of tools we've been discussing and on that on that topic the goal is not to build one tool that does all of this right we could have many tools we could you know different groups could explore a variety of approaches and tools and and uh you know some depths of simplicity and complexity and you know there's room for room for many discussions on on this topic right um one other just just to kind of continue with you the idea of coaching i've been i've been around enough folks in my life to realize that some groups and some people are really skilled at communicating and talking with others and you know getting facilitating group conversation and things like that and and many times i've been just so impressed with the ability of certain groups or certain individuals to to kind of facilitate communication and and show by example what healthy communication looks like and uh and and at the same time i watch tv of you know i've seen tv shows of terrible examples of how to communicate uh you know examples of dishonesty examples of manipulation examples of holding back examples of of confusion you know examples of shallow shallow thinking and uh you know social norms part of this whole process would also be to do like the club model it would be to develop social norms that are useful for the community right that are reinforced to healthy healthy behaviors or healthy ways of communicating health healthy skill development that would help the community to better process the information it has right so so that kind of coaching is available too and this is something in the narrative as active inference we talked about when we see representations of communication like they are up weighted or our priors are updated to think that those types of communication are more likely or that they're prevalent or even that they exist at all and then we fulfill that prediction by reducing our uncertainty about our own behavior and our own thinking through other minds of other people's behavior thinking well what happens when there's two people and they need to is they want the cuts in front and then you're not surprised when they do it you're not surprised when you do it but there's also a way to show representations of that not happening so that it will not surprise either person when they both slow down and that shows how our inter-subjective level of computation is something that is influenced by both the bottom up you know the sensory input and the top down with our cultural scaffolding systems in change need to be able to resonate from the small to the large or there's going to be some um pathologies so stephen and then any other comments so this actually could be an interesting or could inform anyway the types of questions that could compose to carl fristen himself you know around some of these practical implications so we could have these in mind so i was just thinking you know do you have sort of i wonder if you actually have thoughts about oh are there holes in this or are there questions around where this whole field of active inference is sort of going in the sort of global sense and and whether you've thought oh i wonder carl fiston would think about that anyway it's just just a question to put out carl if you're listening feel free to to join in no i would love to i would love to you know we one of the many ways to start this whole process off would be to have a conference where we get together some folks who have experience in these fields in a variety of fields who could offer their thoughts and cautions and encouragements whatever they may be well well john um on june 21st 2021 carl firsten will be joining us for a little symposium and in another sense we have a nano symposium every week but any type of logistics or organization that are the enabling architecture is what active lab is there great fantastic fantastic um fantastic you know i maybe should say too we've been talking about active inference and obviously all of us are interested in that topic but but you know it could be that the active inference 2.0 is called something else it goes you know like we learn something in the next decade or whatever and active inference is no longer you know like we've moved beyond that and now we're talking about some other thing you know what the the important aspect of this for us is how does cognition happen how does you know how how biological organisms uh structure themselves and orient themselves and organize themselves in order to process information and you know how does that work so whether it's active inference or whether it's you know something else sure everything evolves we learn as we go so it might be you know next year it might be a different word but the concept is how do we achieve and maintain vitality into the future you know that's how do we fulfill our purpose and even today for different non-english human languages and for computer languages the word is already not active inference and so that's why it's always so rewarding and valuable to be very clear on what active inference is which parts are drawing from bayesian statistics from complex systems from predictive processing from all of these other areas instead of thinking that we need to reinvent the wheel because we certainly don't we're also just a part of the cognitive architecture and uh it'll be a quite interesting ride for sure so maybe in the last 15 minutes of this point one john help us uh via another slide or by raising another salient question what can we as listeners and participants think about in this next week to um kind of simmer during this week between our two sessions okay um maybe next week we will talk we'll talk more about you know we've we've only covered two slides here we had i think like 60 slides we could have looked at we covered two of them remember maybe three of them uh but the the uh the goal here was to talk about part three of the paper which is design so i think maybe for our listeners maybe the question of like how would you actually do this what would i in parts in part three i i spent some time talking about a a prototype called the what i call the leader framework local economic direct democracy association framework and the emphasis there it is the the concept is a full integrated systems approach but the emphasis in the simulation trial that i published is on economics so just take that alone what would what would a fundamentally different economic system look like how would it function and how how could it apply to a club model you know like implemented at the local level what what what did that look like and when i say economic i mean economic slash financial slash monetary uh institutional you know like all of those things wrapped up into the word economic what how how would that look what are the possibilities we'll go there next maybe great question one thought is um econ eco and uh ecosystem and economy it's like they start with the first three letters they're so close they're almost on the right track with an ecological perspective on economics yet you add one more letter and it changes from uh ecology to economics and all of a sudden it's like never the twain shall meet so it's an awesome question to be rethinking how these different intertwined systems relate just like intertwined systems in an ecosystem of like primary producers prey animals predators renewal etc right right so so on that you know just to follow that up just a little more what is an economic system you know before we go about designing new ones what what is the purpose of an economic system we ought to ask that question first before we jump into potential designs right so what is what what is it supposed to achieve in the first place and the i think for some listeners or for you know the general population at least if you ask that question the answer would be like duh an economic system is about to you know allocating resources back and forth you know choosing who gets what and how resources are used but i would argue that that's not at all with an economic system you know like like intrinsically potentially what an economic system is i would say an economic system is exactly what we've been talking about so far today it is it is a cognitive system it is a it is a different variety of cognitive system that allows a community to cognate in a slightly different way like a governance system allows its community to cognate in one way legal system allows a community to cognate in a different way and an economic system allows it to cognate in a different way so if so if the goal is functional societal cognition what does an economic system look like what one thought on that and using the bodies intertwined signaling systems like neurological chemical etc from the point of view of the ribosome the protein translating machinery insulin is just another peptide it's just linking the amino acids together and it's making a protein but to the point of view of a receptor the insulin molecule is in affordance for binding and then to the point of view of the downstream signaling cascades inside the cell we start to see semantics enter into the picture and then there's a turnover for the insulin and there's lactate being shuttled from the muscles which are using it and they're they're calling out for help through these different mechanisms so it's like a cargo ship might be carrying something and it's putting it in a box and it's just moving it like an encrypted message it's just moving the information it doesn't need the semantics to be able to relay the information and similarly when there's these intertwining systems it's like that one insulin peptide can be handled by a producer by a transporter by a signal detector by a degrading enzyme and the idea that all of those pieces are uh you know you could just change one out or you could remove one it wouldn't be the same system as a whole so stephen and then any other thoughts i like the hormone um reference um as i think that the the nature of the hormone system where you you know the hormone gets released into the blood let's say the blood and it travels around the body it is actually maybe closer to this idea of a message passing kind of signal going out there you know sort of sense in sort of a general way which is and maybe this is something that the economic system does right is and you can't do that so easily through necessarily just dynamical processes with people right it's like okay you can you can kind of infer things you can sort of build things up but okay you set up a pipeline of value distribution or at least the information in relation to value distribution and how that can be processed and you can release your hormones which is basically money budgetary predictions into that system and it flows right it flows around it and it gets picked up and it becomes more like a traditional signal that can be computed and but you can't put out the dynamical system information because it it's not a thing it's done it is dynamical so there's something interesting there i think around that that that analogy to it's the system it's the system we're working with to be able to explore how message passing algorithms instrumentally used by us as curious and motivated investigators allow us to act in the world to reduce our uncertainty about blood sugar so following on that the question two for the public is what is money you know like in its best sense what is money we'll try it we'll try to tackle some of that next week when we meet again yep so just on a closing note we heard two big questions right there at the end how can we actually do it and it would be awesome if people could read especially the third paper in the series and look at the letter framework that john has lita that uh john has proposed and there's a simulation there and it's a scaffold for a lot of rich discussion so on the how can we actually do this we want to hear many perspectives as well as people's thoughts on the third section of paper and then the second question what is money which is of course a fascinating question for history anthropology game theory so many different areas converge on what is money and it's a system in flux as we see quite literally day to day depending on how closely you follow crypto thank you everybody for joining us has been fantastic awesome discussion so everybody who's um watching live thanks so much for participating and next week in number 21.2 we'll have a follow-up discussion with you john so thanks again for your engagement and for really giving us this foundation with the multiple precursor videos because it really helped and i think this is gonna be a great uh conversation sequence so peace out everybody see you later bye thanks bye thanks bye-bye